NATIVE COMPATIBILITY RECORD

Super Glove Ball Native Compatibility

Confirmed behavior, open questions, tracing, and safe fallback operation.

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This document records evidence for the separately named `lr-nestopia-powerglove` core. It intentionally separates observations from hypotheses. Native detection, Start, continuous X/Y, orientation, stale-state safety, and live cabinet control are confirmed for the exact tested ROM. Signed Z and open/fist/index packet bytes are confirmed in exact-ROM headless traces and await live gameplay confirmation. FCEUmm remains the supported explicit fallback using PowerGlove Vision's shared responsive D-pad and gesture recognition.

Evidence order

When sources disagree, use this order:

1. The exact user-supplied Super Glove Ball ROM's input routines and control flow.
2. Controlled emulator traces of its writes, reads, assembled bytes, and cadence.
3. Repeatable in-game detection, out-of-range, and movement behavior.
4. Nestopia's existing Power Glove implementation.
5. The game manual and NESdev reverse-engineering notes.

NESdev material is a source of testable hypotheses, not a specification for this implementation.

Current compatibility status

Detail	Status	Evidence or next test
Shared camera recognition can supply continuous normalized X/Y	Confirmed in application tests	The authenticated receiver publishes the same calibrated axes used by gameplay.
A custom core can consume one coherent latest sample per emulated frame	Confirmed in build and unit tests	Versioned 64-byte read-only record with matching even guards; there is no queue or second smoothing stage.
Missing, uncalibrated, wrong-profile, or older-than-250 ms samples are neutral	Confirmed in implementation tests	The receiver also publishes a neutral record on transport timeout and shutdown.
The candidate core builds separately from stock Nestopia	Confirmed at pinned revision `5a1cd378cb46ca9ccc2dd6f8b2b6a79ab986052e`	Its library name is `Nestopia PowerGlove`; stock source and installed cores are not modified.
Candidate X/Y encoding reaches Nestopia's existing Power Glove device	Confirmed for the exact ROM	Minimum, center, and maximum X/Y each produced distinct packets. Cabinet validation corrected the camera-to-Nestopia Y orientation.
Detection signature, packet length, boundaries, and bit order	Confirmed	The ROM assembled inverse `$A0` as `$5F`, strobed once per byte, read ten bytes/80 bits per sample MSB first, and required the final stored byte to be `$3F`.
Start encoding	Confirmed	Native byte 6 value `$82` left the title screen and began play while the controller stayed in native mode.
Native Z encoding	Packet mapping confirmed headlessly; live gameplay pending	Calibrated camera depth is sign-reversed into the hardware convention. Neutral produced `$00`; maximum forward motion produced `$81`. Test fist plus forward motion as Power Punch on the cabinet.
Native open, fist, and index-point encoding	Packet mapping confirmed headlessly; live gameplay pending	The exact ROM repeatedly received `$00` open, `$FF` fist, and `$0F` index-point samples. Shared five-finger recognition determines compound poses before transmission. Test throw, grab/catch, and Robo-Bullet behavior on the cabinet.
Native wrist rotation and remaining action buttons	Not yet mapped in-game; deliberately neutral	Roll and action recognition are confirmed in the shared layer and FCEUmm output. Vary each remaining native field independently before enabling it. Start is the sole live-confirmed native button.
Poll timing tolerances	Confirmed for tested sessions	Headless runs sustained ten-byte polling throughout native phases, and live cabinet sessions remained stable. Broader hardware and timing stress coverage remains useful.
Headless X/Y activation and release responsiveness	Confirmed for the exact ROM	All four axes visibly diverged by frame 3; a 3.1% positive-X step also diverged by frame 3. See the direction-response benchmark.
Cabinet field mapping and stabilization	Confirmed for live tuning	Continuous X/Y maps each side of the calibrated neutral point to the corresponding usable camera boundary, retaining an 8% tracking margin. Light adaptive damping operates in camera space, reducing near-rest jitter without delaying deliberate travel. FCEUmm D-pad thresholds remain hand-relative and unchanged.
Portable defaults versus neutral calibration	Confirmed in application and installer tests	Full-field mapping, stabilization, and recognition thresholds ship in the release-owned profile baseline. The camera/player-specific neutral reference uses 24 observations at 70% confidence or better and remains private across updates.

Detail	Status	Evidence or next test
Explicit FCEUmm joystick fallback for the same ROM	Confirmed	The ROM enters play using standard Start, requests only the libretro joypad callback, and activates/releases every D-pad direction visibly by frame 3.

The validated ROM is `Super Glove Ball (USA)` with SHA-256 `ad60ef1b62cd1b3bc02a9320376067347a8ab2ebbe46e1616693d8379c9d9a7b`. It remains outside the source tree and release packages. These results apply to that exact image; never silently turn a NESdev assumption into a compatibility claim for another revision.

Confirmed exact-ROM packet

The ROM does not consume the full twelve-byte shape described by some secondary notes. It reads this ten-byte sample, with each transmitted bit inverted by the ROM while assembling the byte:

Byte	Confirmed use	Neutral/test values
0	Detection signature	`$A0`, assembled by the ROM as `$5F`
1	X	`$80` minimum, `$00` center, `$7F` maximum
2	Y	`$80` minimum, `$00` center, `$7F` maximum
3	Signed Z/depth	`$00` neutral; forward camera motion maps toward `$81`; away maps positive
4	Roll candidate	`$00`; deliberately neutral pending exact-ROM mapping
5	Hand gesture	`$00` open, `$FF` fist, `$0F` index point
6	Button	`$FF` neutral; `$82` Start confirmed
7-8	Unknown	`$00`, preserved conservatively
9	Validation terminator	`$3F`

The trace runner starts the exact ROM with the Power Glove attached, proves that native `$82` Start enters play, and holds each X/Y extreme for 120 frames. The captured screens place the Robo-Glove at left, center, right, bottom, center, and top respectively. Separate 60-frame phases then transmit open, fist, open, index-point, open, and fist-plus-forward-Z packets. Tracking-lost, uncalibrated, and stale phases prove that the core returns neutral axes, pose, and buttons instead of retaining the last sample.

Latest-sample interface

The RetroPie receiver owns `/run/powerglove/native-state` and creates it read-only for consumers. Format version 1 is a fixed 64-byte little-endian record containing:

- magic, format version, record size, and matching begin/end coherence guards;
- sample sequence and receiver-arrival monotonic timestamp;

- signed normalized X, Y, Z, and roll axes;
- detected and calibrated flags;
- four compact finger-flex levels;
- recognized-button and compound-pose mask, including five-finger fist and index-point decisions made by the shared recognizer;
- active-profile identifier;
- reserved bytes that stay zero.

The writer publishes an odd in-progress guard and then an even complete guard. The core copies one record at the beginning of its input callback and rejects it unless both guards match and are even. It consumes only current, calibrated, detected Super Glove Ball samples. Stale or invalid input leaves the emulated device neutral.

Build the research core

The build script clones the official libretro Nestopia repository at the pinned revision into a dedicated build directory, applies the local patch, and emits `nestopia_powerglove_libretro.so` without changing a stock installation:

```sh
scripts/build-nestopia-powerglove.sh
```

The normal RetroPie installer offers this source build when it finds a registered Super Glove Ball ROM. Accepting installs Git and standard build tools, builds in a temporary directory, installs the core under its separate name, copies the upstream GPLv2 `COPYING` file beside it, and adds the native entry to the launch menu. It deliberately leaves that ROM's current FCEUmm selection unchanged.

Set `POWERGLOVE_NATIVE_STATE` to use a test record at a different path. Set `POWERGLOVE_TRACE=1` when launching the custom core to log controller writes, latch/counter transitions, returned stream bits, and each candidate output packet. Traces may contain gameplay timing but no camera imagery.

Exact-ROM validation gate

The exact-ROM software gate below passes for detection, Start, X/Y, Z and hand pose packet publication, safe neutralization, and the tested X/Y cabinet path. Live grab/throw, index-fire, and Power Punch behavior remains the next acceptance step. Repeat this gate before enabling the per-ROM emulator choice on another cabinet or after changing the core protocol:

1. Record the ROM digest and retain the ROM outside release packages.
2. Trace controller strobes and configuration writes from power-on through the game's detection decision.
3. Prove the detection signature, packet boundary, bit order, and polling cadence from those traces.
4. Hold every field neutral, then vary X, Y, and Z independently through minimum, center, and maximum values.
5. Transmit open, fist, and index point independently, returning to open between each pose.
6. Confirm repeatable continuous movement plus grab/throw, Robo-Bullet, and fist-plus-forward Power Punch behavior without relying on packet logs alone.
7. Test stale samples, tracking loss, and unavailable calibration; all must immediately yield neutral native input.

8. Build the core on the RetroPie host under the separate name `lr-nestopia-powerglove`, verify the camera-to-receiver path, and only then create the per-ROM override.

Keep an explicit FCEUmm per-ROM choice available. If native detection or tracking regresses, remove only the per-ROM override; the shared FCEUmm fallback remains playable.

Compare both modes from EmulationStation

After the custom core has been installed and registered once, RetroPie's launch menu lists both `lr-fceumm` and `lr-nestopia-powerglove` for this ROM. Choosing `lr-fceumm` keeps the entire session in standard joystick mode: it receives the shared Super Glove Ball profile's D-pad, A, B, Start, and Select outputs and does not read the native-state bridge. The native entry attaches device `517` and reads continuous coordinates instead.

The launch-menu selection for a ROM is persistent, so a test session should be followed by choosing `lr-nestopia-powerglove` again if that is the desired saved default. The command-line selector below performs the same reversible per-ROM choice; neither route changes the system-wide NES emulator.

On the RetroPie host, build/install and then opt in one exact ROM:

```sh
SH
sudo scripts/install-nestopia-powerglove.sh
sudo python3 scripts/configure-super-glove-ball-core.py \
  --rom "/home/pi/RetroPie/roms/nes/Super Glove Ball (USA).7z" \
  --mode native --apply
```

The selector writes the libretro device value `517` and also supplies `--device=1:517` to RetroArch. The explicit launch option ensures this RetroArch build attaches the Power Glove to port 1 before the ROM performs its detection poll. Roll back only that ROM at any time:

```sh
SH
sudo python3 scripts/configure-super-glove-ball-core.py \
  --rom "/home/pi/RetroPie/roms/nes/Super Glove Ball (USA).7z" \
  --mode fceumm --apply
```

To add the remaining already-recognized controls to the native path, vary wrist rotation and one button field at a time. Repeat the neutral baseline, single-variable trace, and observable in-game comparison. Preserve unknown bytes and timing behavior conservatively.